

DeepFake Audio Detection: Methods, Evaluation, and Networked Security Implications

Divyansh Garg

Department of Computer Science and Engineering

Madan Mohan Malaviya University of Technology, Gorakhpur, India

Email: divyanshgarg515@gmail.com

Abstract—The exponential progress of neural text-to-speech (TTS) and voice conversion (VC) technologies has led to the creation of highly realistic synthetic voices, known as deepfake audio. These advancements, while enabling accessibility and creativity, have simultaneously created new threat landscapes in telephony, voice authentication, and digital communication systems. This paper presents an in-depth exploration of deepfake audio detection methods, emphasizing the intersection between machine learning-based detection and network-level security integration. It details the evolution of feature extraction techniques, contrasts CNN and Transformer-based architectures, and proposes an evaluation protocol robust to channel variations and telephony distortions. Furthermore, it introduces a conceptual framework for fusing audio-based detectors with network telemetry (SIP signaling, RTP anomalies, IP reputation) to enhance real-world defense mechanisms. Comprehensive discussions on adversarial robustness, privacy, ethical challenges, and deployment best practices are provided, concluding with strategic recommendations for future defense architectures in the era of synthetic media.

Index Terms—Deepfake audio, spoof detection, voice anti-spoofing, VoIP security, telephony, convolutional neural networks, transformers, ASVspoof, adversarial attacks.

I. INTRODUCTION

In recent years, deep learning has revolutionized speech synthesis through models like Tacotron, WaveNet, and VITS, enabling artificial voices that are nearly indistinguishable from human speech. However, this technological leap has also fueled malicious use cases. Deepfake audio—synthetic or manipulated speech intended to mislead—can impersonate individuals, bypass voice biometric authentication, or conduct large-scale misinformation campaigns.

Traditional security mechanisms in voice authentication or telephony systems, such as automatic speaker verification (ASV), are ill-equipped to distinguish these advanced synthetic signals. Attackers can exploit publicly available TTS or VC frameworks, transmit their deepfakes through VoIP or SIP-based networks, and cause financial or reputational damage. This scenario necessitates a comprehensive, channel-robust, and scalable defense approach.

This research makes three key contributions:

- 1) A detailed review of deepfake audio detection methods, covering spectral, temporal, and learned feature approaches with CNN and Transformer architectures.
- 2) A reproducible evaluation framework that accounts for telephony distortions, codec variability, and adversarial channel noise.

- 3) A network-integrated defense paradigm that fuses deep learning-based spoof detection with network telemetry, thus bridging machine perception and network security.

II. BACKGROUND AND RELATED WORK

A. Deepfake Audio and Attack Modalities

Deepfake audio can originate from several synthesis techniques:

- 1) **Text-to-Speech (TTS)**: Converts textual input into synthetic speech using neural vocoders (e.g., WaveGlow, HiFi-GAN). Produces high-fidelity speech that mimics target voice timbre.
- 2) **Voice Conversion (VC)**: Transforms the spectral features of one speaker's voice into another's while preserving linguistic content.
- 3) **Replay Attacks**: Employ pre-recorded genuine audio samples played back through channels to impersonate a user.

Each technique introduces distinct artifacts—temporal discontinuities, unnatural phase transitions, or high-frequency inconsistencies—which form the foundation for feature-based detection.

B. Benchmarks and Datasets

The **ASVspoof** challenge series (2015, 2017, 2019, 2021) provides standardized datasets for training and evaluating spoof detection systems. These datasets include both bonafide and spoofed utterances across multiple attack algorithms and recording conditions.

For real-world relevance, evaluation under degraded channels—such as AMR, GSM codecs, VoIP jitter, and environmental noise—is critical. Additional corpora such as *FakeAVCeleb*, *WaveFake*, and *In-the-Wild DeepFake Audio* provide complementary perspectives for generalization testing.

C. Feature-Based and Learned Detection Methods

Early countermeasures relied on handcrafted features:

- **CQCC (Constant-Q Cepstral Coefficients)**: Captures frequency-dependent resolution of audio.
- **LFCC/MFCC**: Linear or Mel-frequency cepstral representations reflecting perceptual properties.

Modern systems employ deep neural networks trained on spectral features (log-mel spectrograms) or directly on raw

waveforms. CNNs excel in capturing local spectral patterns, whereas Transformer-based models leverage self-attention to identify long-term dependencies and synthesis inconsistencies.

III. PROBLEM FORMULATION

We define the problem as binary classification:

Given an audio segment x , predict its label $y \in \{\text{bonafide}, \text{spoof}\}$. A trained model $f(x; \theta)$ outputs a score $s = f(x) \in [0, 1]$, representing spoof likelihood. A threshold τ converts s into a binary decision.

$$\hat{y} = \begin{cases} \text{spoof}, & s \geq \tau \\ \text{bonafide}, & s < \tau \end{cases}$$

Performance metrics include Equal Error Rate (EER), Receiver Operating Characteristic Area Under Curve (ROC-AUC), precision, recall, and Detection Cost Function (DCF) when coupled with ASV systems.

IV. METHODS

A. Preprocessing and Feature Extraction

1) Audio Normalization and Resampling: All inputs are resampled to 16 kHz. Energy-based voice activity detection (VAD) removes silence regions, ensuring temporal compactness.

2) Spectral Representation: The Short-Time Fourier Transform (STFT) with 25 ms windows and 10 ms hop size is computed. The magnitude is converted to log-mel spectrograms with 128 mel bins. Per-utterance normalization (zero mean, unit variance) enhances generalization.

3) Data Augmentation: Additive noise (SNR: 5–30 dB), reverberation (RT60: 0.2–0.6s), codec compression (AMR, G.711, Opus), and speed perturbation ($\pm 10\%$) simulate real-world telephony artifacts. This improves channel robustness and adversarial resilience.

B. Model Architectures

1) Convolutional Neural Network (CNN): A compact CNN architecture processes 2D log-mel spectrograms:

- Four convolutional layers with kernel sizes (3×3), each followed by batch normalization and ReLU.
- Global average pooling aggregates temporal information.
- Fully connected layer with sigmoid activation outputs spoof likelihood.

This model captures fine-grained spectral irregularities, making it efficient for on-device inference.

2) Transformer-Based Model: Spectrograms are patchified (e.g., 16×16) and linearly projected into tokens. These are fed into a transformer encoder comprising:

- Multi-head self-attention (8 heads) to capture global dependencies.
- Position embeddings to preserve temporal order.
- Feed-forward layers with GELU activation for non-linearity.

Transformers excel at modeling cross-frame relationships, essential for detecting subtle synthesis artifacts.

3) Ensemble Framework: Ensemble averaging between CNN and Transformer predictions leverages their complementary strengths—CNN for local frequency analysis and Transformer for global context modeling.

C. Training Configuration

The training employs binary cross-entropy loss with label smoothing ($\epsilon = 0.1$). Optimization uses AdamW (lr = $3e-4$, weight decay = $1e-5$) with cosine annealing and warmup. Early stopping monitors validation EER. Training is conducted for 50 epochs with batch size 32.

V. EXPERIMENTAL EVALUATION

A. Dataset Setup

Experiments utilize ASVspoof 2019 LA subset. Training includes multiple synthesis algorithms; testing introduces unseen attacks. A 20% validation split ensures unbiased evaluation. For robustness testing, telephony codecs (AMR-NB, GSM) and additive noise are applied to test samples.

B. Evaluation Metrics

Primary metrics:

- **Equal Error Rate (EER):** Point where false acceptance equals false rejection.
- **ROC-AUC:** Area under receiver operating curve.
- **Detection Latency and Model Size:** Key deployment factors for telephony gateways.

C. Baseline Models

- CQCC + GMM:** Classical baseline using handcrafted features.
- CNN Model:** Trained on log-mel inputs.
- Transformer Model:** Attention-based spectral modeling.
- Ensemble:** Weighted averaging of CNN and Transformer predictions.

D. Results and Analysis

TABLE I
PERFORMANCE COMPARISON ON CLEAN AND TELEPHONY-DEGRADED AUDIO

Model	EER (Clean)	EER (Telephony)
CQCC + GMM	10.0%	24.3%
CNN (log-mel)	3.2%	8.5%
Transformer	2.7%	7.1%
Ensemble (CNN+Transformer)	2.4%	6.4%

The Transformer demonstrates superior robustness under telephony distortions due to its global context modeling. The ensemble further improves average performance, confirming that hybrid architectures mitigate overfitting to specific artifacts.

VI. NETWORKED SECURITY IMPLICATIONS

A. Threat Landscape and Delivery Channels

Voice-over-IP (VoIP) and SIP-based telephony allow remote, automated delivery of deepfake voices. Attackers can integrate neural vocoders into botnets or call centers, amplifying social engineering attacks at scale.

B. Fusing Audio Detection with Network Telemetry

Detection accuracy improves significantly when audio-based inference is correlated with network-level indicators:

- **SIP Header Anomalies:** Inconsistent User-Agent or malformed fields.
- **Transport Features:** Suspicious IPs, geolocation mismatches, or latency anomalies.
- **RTP Media Indicators:** Irregular packet sizes, jitter spikes, or codec inconsistencies.

A multi-layer correlation engine fusing these indicators reduces false positives and supports real-time mitigation.

C. Deployment Architecture

A conceptual architecture integrates detection into the telephony stack:

- Audio snippets (3–10s) are extracted at the ingress Session Border Controller (SBC).
- The snippet is passed to the deepfake detector for inference.
- Results are forwarded with SIP metadata to a Security Information and Event Management (SIEM) system.
- Automated policies trigger call blocking, challenge-response authentication, or human review.

VII. DISCUSSION

A. Adversarial Robustness

Attackers may employ adversarial post-processing (e.g., pitch shifting, time-stretching, or channel equalization) to fool detectors. Countermeasures include:

- Adversarial data augmentation.
- Regular model retraining on newly observed attacks.
- Defensive distillation and uncertainty estimation to detect out-of-distribution samples.

B. Ethical and Privacy Considerations

Deepfake detection inherently involves processing biometric data. Hence, privacy-preserving pipelines are crucial:

- Avoid storing raw audio; retain only intermediate embeddings.
- Comply with GDPR and data minimization principles.
- Ensure model interpretability to justify false positives.

C. Limitations

Although this framework outlines a robust design, real-world telephony introduces unpredictable variations—background noise, codec cascades, and latency—which may still degrade performance. Continuous retraining and adaptive threshold tuning are essential.

VIII. RECOMMENDATIONS AND BEST PRACTICES

- 1) Train detectors on channel-augmented, codec-varied datasets for improved generalization.
- 2) Deploy hybrid CNN-Transformer ensembles to capture both local and global audio cues.
- 3) Integrate audio detection results with SIP/RTP telemetry for holistic verification.
- 4) Position detectors at the ingress gateway for low-latency analysis.
- 5) Include a human-in-the-loop escalation path for ambiguous cases.

IX. CONCLUSION

Deepfake audio poses a multidimensional security risk—technical, social, and operational. Reliable countermeasures demand not only powerful models but also an integrated defense framework uniting AI-based detection with networked intelligence. Through comprehensive modeling, adversarial robustness, and telephony-aware design, systems can evolve toward trustworthy voice communication in a deepfake-saturated world.

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